BRD: QuickEntry for Tally (Offline Field Sales App)

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1. Summary

QuickEntry for Tally lets field salespeople create sales invoices, receipts and sales orders on a phone, tablet or PC, with or without internet, and posts them into TallyPrime without duplicates. TallyPrime stays the accounting system and the source of all customer, supplier and item masters; the app is a fast front end for staff outside the office.

Item	Detail
Document	Business Requirements Document (BRD), version 0.1 draft for client review
Product	QuickEntry for Tally, an installable web app (PWA) plus a small office connector for TallyPrime
Client context	UAE business on TallyPrime; AED currency; VAT 5% today, but VAT must be configurable
Status	Draft. Open questions in section 10 need client answers before the design is final

Business goals

1. Remove double entry: a transaction typed once in the field reaches Tally without re-keying.
2. Keep salespeople productive with no signal: create, save, print and share documents offline.
3. Protect the books: no duplicate vouchers, no lost work, and a clear status for every transaction.
4. Fewer errors: customers, items, rates and VAT come from Tally masters, not free typing.

2. Users and current state

TallyPrime runs on a PC in the client's office and holds all ledgers, stock items and vouchers. Today field staff write orders and invoices by hand or in messages, and office staff re-key them into Tally.

User	Needs from the app
Salesperson (field)	Find a customer and items in seconds, enter quantity and rate, save and share an invoice, receipt or order with no signal
Office / accounts staff	See what came from the field, fix failed syncs, keep Tally masters correct
Manager / owner	Daily totals, confidence that nothing is duplicated or lost, control over who can do what
Purchase staff	Record purchase transactions (workflow to be confirmed, question Q4)

What exists today (inspected for this BRD). A clickable prototype, QuickEntry for Tally, shows the six entry screens, a list with Pending, Synced and Failed badges, and PDF and WhatsApp sharing. It is a design reference only: data lives in memory and is lost on reload, records are matched by name, VAT is fixed at 5%, and Tally sync is simulated. The production build keeps its screen flow and replaces everything underneath.

3. Scope

Release 1 covers field sales (invoice, receipt, order) end to end, plus purchase entry once its workflow is confirmed.

In scope (release 1)	Out of scope
Installable app for phone, tablet and desktop (PWA)	App Store / Play Store builds (possible later from the same code)
Sync of customer, supplier, bank/cash ledgers, stock items, units and VAT classes from Tally	Creating or editing masters in the app (masters are maintained in Tally only)
Sales Invoice, Receipt, Sales Order screens	Payroll, manufacturing, job work, cost centres
Purchase transaction screen (exact workflow per Q4)	Reports beyond a simple daily summary and transaction list
Offline save, view and edit; automatic and manual sync	UAE e-invoicing (Peppol) submission, unless confirmed in Q10

In scope (release 1)	Out of scope
One-way posting of app transactions into Tally, with update of the same voucher on edit	Pulling vouchers keyed in Tally back into the app (only status and conflicts are read back)
PDF, print and WhatsApp share of invoices and receipts	Payment collection (card links) and customer self-service
Configurable VAT rates and calculation rules	Hardcoded jurisdiction logic
Office connector program on the Tally PC	Tally ERP 9 support, unless confirmed in Q1
A demo mode with sample data and a simulated Tally	

The structure supports more voucher types later (Credit Note, Payment, Purchase Order) by adding a definition, not new plumbing.

4. Business requirements

Each requirement has an ID for tracing to tests; Must items are needed for go-live.

ID	Requirement	Priority
Devices and usability		
BR-01	Works on phones, tablets and desktop browsers; installs to the home screen like an app	Must
BR-02	Large touch targets, plain words, few fields per screen, for users with limited technical knowledge	Must
BR-03	Customer and item search by name, code, phone or TRN returns results as the user types, with 10,000+ items and 5,000+ customers stored on the device	Must
BR-04	Quantity and rate entry with a numeric keypad; rate pre-filled from the item master and editable if allowed	Must
Master data		

ID	Requirement	Priority
BR-10	Ledgers (customers, suppliers, bank/cash) and stock items are created and maintained in Tally only	Must
BR-11	Every master has a stable, unique code used to match it across systems; names are never used as identifiers	Must
BR-12	A rename or other change in Tally updates the existing app record at the next sync, matched by code	Must
BR-13	First sync is a full load; later syncs transfer only new or changed masters	Must
BR-14	Masters deleted or deactivated in Tally are hidden in the app, not lost from past transactions	Should
Transactions		
BR-20	Sales Invoice: customer, items, quantity, rate, discount, VAT per line, totals, narration	Must
BR-21	Receipt: customer, amount, mode (cash, bank, cheque, PDC), reference, invoices settled	Must
BR-22	Sales Order: customer, items, quantity, rate, delivery date	Must
BR-23	Purchase transaction: screen built to the workflow confirmed in Q4	Must (after Q4)
BR-24	New transaction types can be added by configuration (voucher type, fields, Tally mapping)	Should
Offline		
BR-30	Create, save, view and edit transactions with no internet	Must
BR-31	Masters and transactions are stored on the device and survive app close, restart and failed syncs	Must
BR-32	At least the last 2 days of transactions stay on the device for view and edit; unsynced ones are kept regardless of age	Must

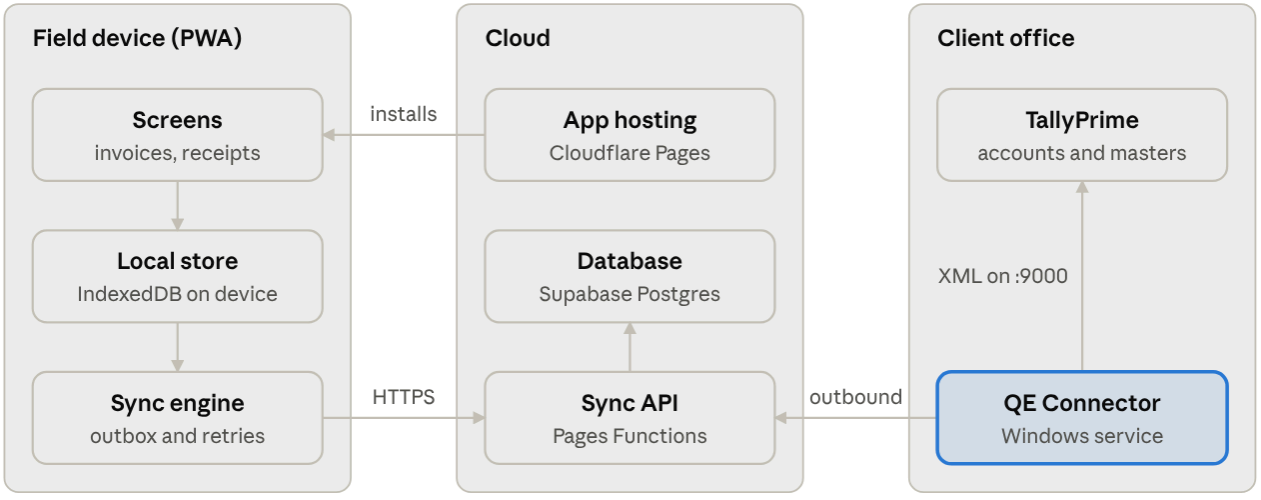
ID	Requirement	Priority
BR-33	Sync starts automatically when internet and the Tally connection are available; a Sync now button is always available	Must
BR-34	Every transaction shows Pending, Synced, Failed (with reason) or Conflict	Must
Sync with Tally		
BR-40	Every transaction has a permanent unique ID, mapped to its Tally voucher	Must
BR-41	Repeating a sync, or re-sending the same records, never creates duplicate vouchers	Must
BR-42	Editing a synced invoice or receipt updates the same Tally voucher at the next sync	Must
BR-43	Only new or changed transactions are sent after the first transfer	Must
BR-44	Interrupted transfers retry safely; errors are logged with Tally's reason and can be retried	Must
BR-45	A change made to the same voucher in both Tally and the app is detected and handled by the agreed conflict policy (Q6)	Must
BR-46	The cloud never connects into the office network; the office connector connects out	Must
Documents and VAT		
BR-50	Invoice and receipt documents show company details, TRN, quantities, rates, VAT and totals	Must
BR-51	Export to PDF, print, and share to WhatsApp through the device share sheet, including offline	Must
BR-52	VAT rates, rounding and inclusive/exclusive pricing are settings, not code; VAT class comes from the Tally item or ledger	Must
Security		
BR-60	Each user signs in; data is separated per company (tenant)	Must

ID	Requirement	Priority
BR-61	Tally connection details and connector keys stay in the office and the server, never on field devices	Must
BR-62	User roles limit which transaction types each user can create (per Q3)	Should

5. Proposed solution

Three parts, each replaceable on its own: the app on the field device, a small cloud service, and a connector on the office PC next to Tally.

Field devices and Tally never talk directly; the cloud relays



The connector polls out over HTTPS; nothing in the cloud opens a connection into the office.

solution architecture · device, cloud, office

- **App (PWA).** Screens, a local database on the device (IndexedDB), and a sync engine with an outbox. Everything a salesperson does is saved locally first, so it works with no signal.
- **Cloud.** Hosts the app and a small sync API in front of a Postgres database. It holds each company's masters, a copy of every transaction, the transaction-to-voucher mapping and the sync log. Each company's data is isolated by row-level security.
- **Office connector.** A small Windows service on the Tally PC. It reads changed masters from Tally and posts pending transactions into Tally through Tally's built-in XML/HTTP interface on port 9000, which it reaches only on the local machine. It connects out to the cloud, so no ports open on the office router.

- **Why not connect straight to Tally?** Tally sits inside the office network and is often off at night. The cloud relay lets field devices sync any time, and the connector catches up when Tally is open.

Technology choices. Plain web technologies (HTML, JavaScript, IndexedDB, service worker) for the app, so it installs on any phone without app stores and runs offline. Cloudflare Pages for hosting and API functions (confirmed by the client) and Supabase (managed Postgres with authentication and row-level security) for data, because both have free tiers while building and low fixed prices in production. Node.js for the connector, packaged as a single Windows program, because it runs the same sync code as the cloud and app.

Data flow: masters (Tally to app)

1. The connector asks Tally for ledgers and stock items whose AlterID is higher than the last one it saw (the first run asks for all).
2. It sends them to the sync API, keyed by each master's stable code and Tally GUID; the API inserts new rows and updates changed ones, so a rename overwrites the old name on the same record.
3. Each device asks the API for masters changed since its own last sync and stores them locally, so search stays instant and offline.
4. Once a day the connector sends the full list of codes, so masters deleted in Tally are marked inactive.

Data flow: transactions (app to Tally)

1. The salesperson saves a transaction. It gets a permanent ID (UUID) and a version number on the device and shows Pending.
2. When online, the sync engine uploads pending transactions. The API stores them by UUID, so the same upload twice changes nothing.
3. The connector collects queued transactions and posts each one to Tally carrying its UUID. If a voucher with that UUID already exists, it is altered, not created again.
4. Tally's reply (voucher ID or error) goes back to the API and then to the device: Synced, Failed with Tally's reason, or Conflict.
5. An edit raises the version number and repeats steps 2 to 4 against the same voucher. Before altering, the connector checks the voucher was not changed in Tally since the last sync; if it was, the conflict policy applies.

6. Implementation and hosting plan

The whole system runs on two managed services plus one program in the client's office, for about **USD 41 a month (about AED 150)** in production for the first client, and close to zero while building.

Where each part runs

Component	Runs on	How it is deployed	Who pays
App (PWA)	Cloudflare Pages global CDN, HTTPS	Push to GitHub; Cloudflare Pages publishes automatically (no build step)	You (subscription)
Sync API	Cloudflare Pages Functions (serverless)	Same deploy as the app	You (Workers free or Paid plan)
Database, user sign-in, backups	Supabase managed Postgres	Schema scripts run on each release	You (subscription)
Office connector	Client's Tally PC, Windows 10/11	Signed installer, runs as a Windows service, auto-starts	You (code-signing certificate)
TallyPrime	Client's PC	Already installed; one setting enabled (section 8)	Client (their Tally licence)
Field devices	Staff phones, tablets, PCs	Open the app link, tap Install	Client

Environments. Three, so the client's live books are never used for testing: Development (free tiers, sample Tally data), Staging (pilot with the client's test company in Tally) and Production. Every release goes to Staging first.

Operations built in from day one

- Secrets (database keys, connector keys) live only in Cloudflare Pages environment settings and in the connector's protected config on the office PC; never in the app or in source code.
- Daily database backups (7 days kept on Supabase Pro), plus Tally's own backup, which stays the client's responsibility.
- Free uptime monitoring on the sync API, and an alert when a connector has not checked in for 30 minutes during office hours.

- The connector checks for a newer version on start and can be updated remotely; the app updates itself on next open.

Your monthly subscription costs (USD; prices checked 31 Oct 2026 on the Cloudflare Workers and Pages and Supabase pricing pages; domain and code-signing are approximate)

Service	While building	Production, first client	Notes
Cloudflare Pages (app hosting)	0	0	Free plan; static files and bandwidth are not metered
Cloudflare Pages Functions (sync API)	0 (100,000 requests a day)	5 (Workers Paid, 10 million requests a month)	Free covers about 15 to 20 active users at the default 20-second auto-sync; budget the USD 5 plan for go-live
Supabase database and auth	0 (Free; pauses after a week idle)	25 (Pro, includes one Micro compute)	Includes daily backups and no pausing
Domain name	0 (pages.dev address)	about 1 (.com, about 10 to 12 a year via Cloudflare Registrar)	A .ae domain costs more, about 40 to 70 a year (approximate)
Code-signing for the connector	0 (unsigned; Windows shows a warning)	about 10 (Microsoft Artifact Signing)	Alternative: an OV certificate, about 200 to 400 a year (approximate)
Email for sign-in links	0	0 (free tier of a mail service such as Resend)	Only needed if staff sign in by email link
Monitoring and error logs	0	0 (free tiers)	Upgrade only if volume grows
Total per month	**about 0**	**about 41**	About AED 150 at 3.67 AED per USD

As you add clients. All clients share one database with row-level security, so each new client company adds roughly nothing until usage grows; the next step up is Supabase Small compute, about USD 15 more a month. If a client insists on a separate database, add about USD 10 a month per client.

If the client requires data to stay in the UAE. Supabase has no UAE region today, so data would sit in a nearby region such as Mumbai or Frankfurt. If UAE residency is required (question Q9), the database moves to Postgres on AWS Middle East (UAE) or Azure UAE North, estimated at USD 40 to 80 a month (not price-checked), and the total rises to roughly USD 60 to 100 a month.

7. Data model

The same six records exist on the device and in the cloud; every row carries the company (tenant) ID, and nothing is ever matched by name.

Record	Key fields	Purpose
Ledger master	code (stable, unique), tally_guid, tally_alter_id, name, group (customer, supplier, bank, cash), TRN, phone, email, address, VAT registration type, active, updated_at	Customers, suppliers, bank and cash accounts from Tally
Item master	code, tally_guid, tally_alter_id, name, part_no, unit, VAT class, standard sale and purchase rate, active, updated_at	Stock items from Tally
Transaction	id (UUID made on the device), type (sales invoice, receipt, sales order, purchase), number, date, party code, narration, mode and reference (receipts), totals, version, created_by, device_id, sync_status, updated_at	One business document, whatever its type
Transaction line	transaction id, line no, item code, quantity, unit, rate, discount, VAT rate, VAT amount, amount	Item rows of an invoice, order or purchase
Sync mapping	transaction id, tally voucher GUID, tally master ID, tally voucher number, synced version, tally alter ID at sync, synced_at	Links each app transaction to exactly one Tally voucher
Sync log	transaction or master ID, direction, attempt no, started_at, result (ok, failed, conflict), Tally message	History for troubleshooting and retry

Record	Key fields	Purpose
Settings	VAT rates and rounding, inclusive or exclusive pricing, number series, document header and logo, retention days	Per-company configuration, no hardcoded rules

Sync status values. Draft (not yet saved), Pending (waiting to send), Sent (in the cloud, waiting for Tally), Synced, Failed (with Tally's reason, retryable), Conflict (needs a decision).

Transaction types are data. A type definition lists its Tally voucher type, which fields it shows, its party group and how it maps to Tally XML. Adding Credit Note or Payment later means adding a definition and a mapping, not changing the sync engine.

8. Tally configuration and office setup

The connector uses TallyPrime's built-in XML over HTTP interface, which needs one setting and an open company; a small TDL add-on is recommended for stable codes.

What must run in the office

- A Windows 10 or 11 PC with TallyPrime open and the target company loaded during the hours vouchers should post. When Tally is closed, transactions wait safely in the cloud and post when it reopens.
- The QuickEntry connector, installed as a Windows service on that same PC, with internet access (outbound HTTPS only).

TallyPrime settings

Setting	Where in TallyPrime	Value
XML/HTTP interface	F1 Help, Settings, Connectivity, Client/Server configuration	TallyPrime acts as: Both (or Server); Port: 9000
Inventory in vouchers	F11 Features, Inventory	Maintain stock items: Yes
Sales orders	F11 Features, Inventory	Enable order processing: Yes
VAT	F11 Features, Taxation	Enable VAT: Yes, with the company's TRN and rates
Part numbers (optional)	F11 Features, Inventory	Enable part numbers: Yes

Setting	Where in TallyPrime	Value
Required ledgers	Gateway, Create, Ledger	Sales and purchase ledgers, Output and Input VAT ledgers, a round-off ledger, cash and bank ledgers (exact names to be agreed, Q7)

Connector settings (stored encrypted on the office PC, never on phones): Tally address (http://localhost:9000), exact Tally company name, cloud API address, and a connector key issued per company. Port 9000 must stay closed to the internet; the Windows firewall should block it from other machines unless another office PC needs it.

Stable master codes. Tally does not give masters a business code by default, so the recommended approach is:

Option	How it works	Customization	Verdict
A. Tally GUID only	Use Tally's internal GUID, which survives renames	None	Fallback key. Not readable by people, and changes if the company is rebuilt by export and import
B. Built-in fields	Part No. for items, Alias for ledgers	None	Workable, but staff can edit or clear them, and aliases also appear in name search
C. QE Code field (recommended)	A small TDL adds a QE Code field to ledgers and stock items. The connector gives any master without a code the next number (for example C000123, I000045) and writes it back to Tally	Small TDL file, loaded via F1, TDLs and Add-ons	Recommended. Readable, never reused, survives renames; GUID is stored too as a cross-check

If the client already keeps codes (for example item codes in Part No.), the connector adopts them instead of generating new ones (Q2).

Changes only. Every Tally master and voucher carries an AlterID that rises on each change. The connector remembers the highest AlterID it has sent and asks only for higher ones; this needs no add-on.

No duplicate vouchers. Each voucher is sent with the app's UUID as its remote ID, and the same TDL stores it in a QE Txn ID field. Before creating, the connector looks up that ID in Tally; if found, it alters that voucher. The exact Tally behaviour on re-import must be verified on the client's Tally version during the pilot; until then it is labelled not verified.

9. Assumptions, constraints and risks

The build proceeds on these assumptions; each one can be changed when the client answers section 10.

Assumptions

ID	Assumption
A1	One TallyPrime company per client in release 1, on TallyPrime 4.x or later
A2	Sales Invoice posts as a Tally Sales voucher in item invoice mode; Receipt as a Receipt voucher; Sales Order as a Sales Order voucher
A3	Prices are entered VAT-exclusive; VAT is calculated per line at the item's rate (5% standard by default) and rounded to 2 decimals per line
A4	Invoice numbers are generated on the device as a per-user series (for example SI-JOHN-0001) so two offline users never clash; Tally keeps its own voucher number alongside
A5	Conflict policy until confirmed: Tally wins. If a voucher was changed in Tally after the last sync, the app's edit is not posted and is flagged Conflict for the office to resolve
A6	Synced transactions older than 2 days are removed from the device but remain in the cloud and Tally; unsynced ones are never removed
A7	All stock goes from one default godown; no batches or serial numbers

ID	Assumption
A8	Purchase entry is built as a Purchase voucher (supplier invoice) until Q4 is answered
A9	Masters are read-only in the app; new customers are created in Tally first

Constraints

- Vouchers post only while Tally is open on the office PC; the app always works, but Synced status waits for Tally.
- iPhones clear stored data of web apps not opened for several weeks unless the app is installed to the home screen; staff must install it. The app warns before anything unsynced could be at risk.
- WhatsApp sharing uses the phone's share sheet, so the user picks the chat; the app cannot send messages by itself.

Risks

Risk	Effect	Mitigation
Tally's behaviour on re-importing a voucher differs by version	Possible duplicate or failed alter	Look up by QE Txn ID before every create; test on the client's exact version in the pilot
Ledger or item missing or renamed in Tally while a transaction waits	Voucher fails	Match by code, not name; show Tally's error; retry after the master is fixed
Office PC off for days	Field work does not reach the books	Connector heartbeat alert to the manager
A lost or stolen phone	Exposure of customer data	Sign-in required, sessions revocable, data limited to that user's company

10. Open questions for the client

These answers change the design; the build continues on the assumption shown until each is answered.

ID	Question	Assumed until answered
Q1	Which TallyPrime version and release is in use? Is Tally ERP 9 needed? How many Tally companies?	TallyPrime 4.x or later, one company, no ERP 9
Q2	Do ledgers and items already have codes in Tally (Part No., alias, or in the name)? May we install a small TDL add-on that adds a QE Code field?	TDL allowed; codes generated by the connector
Q3	How many app users, and should roles limit what each can do (for example sales staff cannot record purchases or change rates)?	Two roles: salesperson and office admin
Q4	Purchase workflow: who records purchases, from where (field or office), and is it a supplier invoice, goods receipt, or purchase order? Must stock be received against a PO?	Purchase voucher entered from a supplier invoice
Q5	May salespeople change the rate or give a discount? Within what limit?	Rate editable, discount allowed, no limit
Q6	Conflict policy: if a synced invoice is edited in both Tally and the app, which wins? Should edits be blocked once a voucher is synced, or after a time limit?	Tally wins; app edit flagged Conflict
Q7	Exact names of the sales, purchase, VAT, round-off, cash and bank ledgers and voucher types to post to	Created to match the sample data; to be mapped in settings
Q8	Numbering: should app invoice numbers be the Tally voucher numbers, or may each user have a series?	Per-user series on the device
Q9	Must data be hosted inside the UAE?	No; a nearby Supabase region
Q10	Is UAE e-invoicing (Peppol) submission needed in this project?	Not in release 1
Q11	Are stock levels needed on the device, and are godowns or batches used?	Stock shown from last sync; one godown
Q12	Can you share a current printed invoice and receipt so the PDF layout matches?	Standard UAE tax invoice layout

11. Acceptance criteria and delivery plan

Go-live requires all ten criteria to pass on the client's own Tally company in the pilot, not only in the demo.

#	Acceptance criterion	How it is shown
1	A salesperson creates an invoice and a receipt with no internet	Phone in airplane mode; both saved and shown Pending
2	Offline records remain after restarting the app	Close the app, restart the phone, reopen: records still there
3	Pending transactions sync when connectivity returns	Turn internet back on: status moves to Synced; vouchers visible in Tally Day Book
4	Repeated syncs create no duplicates	Press Sync now several times and re-send a day's data; voucher count in Tally unchanged
5	Editing a synced transaction updates the same Tally voucher	Change a quantity; same voucher number and ID in Tally shows the new amount
6	Renaming a master in Tally updates the app's record by its code	Rename a customer in Tally; after sync the app shows the new name and past invoices still link to it
7	Later master syncs transfer only additions and changes	Sync log shows the number of masters sent: all on first run, only the changed ones after
8	Recent transactions can be viewed and edited on the device	Last 2 days listed offline and editable
9	PDF, print and WhatsApp share work	Generate a PDF, print it, share it to a WhatsApp chat from the phone
10	Customer and item search stays responsive at realistic volume	Search returns in under 100 ms with 5,000 customers and 10,000 items on a mid-range phone

Delivery plan

Phase	What is delivered	Gate to move on
0. BRD and demo (now)	This BRD; a working demo with sample data and a simulated Tally, clearly marked as not live	Client reviews BRD and answers section 10
1. Build	Production app, cloud sync API and database, office connector, TDL add-on, PDF layouts to the client's sample	All ten criteria pass against a test company in Tally
2. Pilot	Two or three salespeople use it on a copy of the client's company, then on live data	One week with no duplicates, no lost entries, failures explained
3. Go-live	All users, production hosting, monitoring and backups switched on	Client sign-off
4. Later (optional)	More voucher types, store apps, e-invoicing, UAE hosting	Separate approval

Sign-off

Role	Name	Date
Client representative		
Solution provider		